

Use the data and formulae from the general handout sheet for any material properties and constants required.

1. Mark the following true or false

- F i Drag on an aircraft is equal to the summation of the skin friction over its surface
- T ii Internal stresses in a fluid depends on the rate of deformation, not on total deformation
- F iii The principle of conservation of energy does not apply to fluid flows
- T iv The principle of conservation of energy is not needed to describe constant temperature flows in air.
- F v Viscous stresses act only in a stationary fluid
- F vi Pressure acts only in a stationary fluid
- T vii Pressure is the same at point in all directions but viscous stresses are not
- T viii "Inviscid" flow is flow with zero or negligible viscosity

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2. The velocity components of a fluid flow are $u=3\text{ms}^{-1}$; $v=4\text{ms}^{-1}$; $w=12\text{ms}^{-1}$. Show that the resultant speed is 13ms^{-1} . An observer now moves in the x -direction at a speed of 5ms^{-1} thereby changing the frame of reference of the flow. What is the resultant velocity of the flow in the frame of reference of the observer? (Ans -2, 4, 12)

$$\text{(i)} \quad u_o = \sqrt{v_u^2 + v_v^2 + v_w^2} = \sqrt{169} = 13 \text{ (m/s)}$$

$$\text{(ii)} \quad u' = 3 - 5 = -2 \text{ (m/s)}$$

$$v' = 4 \text{ (m/s)}$$

$$w' = 12 \text{ (m/s)}$$

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3. The speed of an aircraft is 450 Knots. Express this in (i) km/hr (ii) miles/hr (iii) m/s(iv) ft/s (**Ans 834, 518, 232, 760**)

$$(i) 450 \times 1.853 \text{ km/hr} = 833.85 \text{ km/hr} = 834 \text{ km/hr}$$

$$(ii) 450 \times 1.853 \text{ km/hr} \times \frac{1}{1.61} \text{ miles/km} = 518 \text{ miles/hr}$$

$$(iii) 450 \times 1.853 \text{ km/hr} \times 1000 \text{ m/km} \times \frac{1}{3600} \text{ hr/s} = 231.625 \text{ m/s} = 232 \text{ m/s}$$

$$(iv) 231.625 \text{ m/s} \times 3.28 \text{ ft/m} = 759.73 \text{ ft/s} = 760 \text{ ft/s}$$

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4. Show that the atmospheric pressure at sea level, $1.013 \times 10^5 \text{ Nm}^{-2}$, is equal to the pressure difference between the top and bottom of a water column of height 10.33m. Consider the density of water as 1000 kg/m^3 and $g = 9.81 \text{ m/s}^2$.

$$\begin{aligned}\Delta P &= \rho gh = 1000 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times 10.33 \text{ m} = 1.013373 \times 10^5 \text{ N/m}^2 \\ &= 1.013 \times 10^5 \text{ N/m}^2 \\ &= P_0\end{aligned}$$

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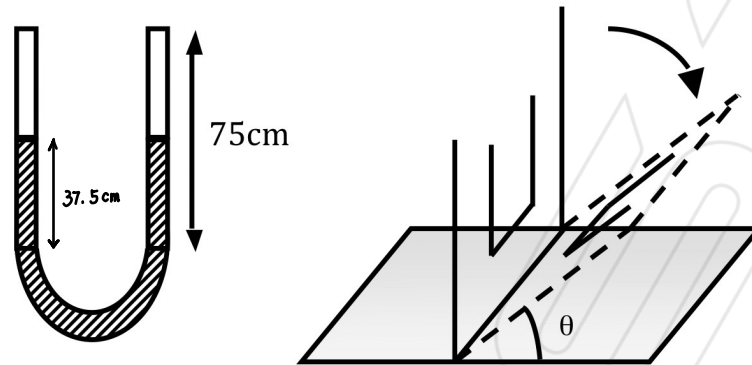


Figure 1

5. As shown in figure 1 a U-tube manometer has two straight sides each of length 75 cm and is filled with alcohol of density 800 kg/m^3 to the *half-way* mark on the scale. If it is to be used to measure the pressure rise produced by a centrifugal blower (claimed by the manufacturer to create a pressure equal to 15.3 cm of water) what is the smallest angle to the horizontal at which the manometer may be inclined without spilling liquid while in use? (**Ans 14.77°**)

$$\Delta P = \rho g h = 1500.93 \text{ N/m}^2$$

$$\Delta h = \frac{\Delta P}{\rho g} = 0.19125 \text{ m} = 19.125 \text{ cm}$$

$$\therefore \frac{\Delta h}{\sin \theta_{\max}} = 75 \text{ cm}$$

$$\therefore \theta_{\max} = 14.774^\circ = 14.77^\circ$$

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6. The pressure at a specific point on the upper surface of an aerofoil is found in a wind tunnel test to be equal to 25cm of water below the value of the atmospheric pressure of 1 bar. What is the pressure in: (i) Pa (N/m^2) (ii) MPa (N/mm^2) (iii) PSI (lb/sq in) (**Ans 97548, 0.0975, 14.14**)

$$\begin{aligned}P &= P_0 + \rho gh = 1 \text{ bar} - 2452.5 \text{ N/m}^2 \\&= 100 \text{ kPa} - 2452.5 \text{ Pa} \\&= 97547.5 \text{ Pa} \\&= 97548 \text{ Pa} \\&= 0.0975 \text{ MPa} \\&= 97548 \text{ Pa} \times \frac{1}{6894.76} \text{ psi/Pa} \\&= 14.15 \text{ psi}\end{aligned}$$